



A.4 Rigid body mechanics (HL)

Space, time and motion · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Rotational motion is described by angular displacement θ , angular velocity $\omega = d\theta/dt$, and angular acceleration $\alpha = d\omega/dt$, analogous to linear kinematics with $v = \omega r$ and $a = \alpha r$ for a point at radius r . The moment of inertia I measures rotational inertia (resistance to angular acceleration) and depends on mass distribution relative to the rotation axis; torque $\tau = I\alpha$ is the rotational analogue of $F = ma$, and angular momentum $L = I\omega$ is conserved when net external torque is zero. Rotational kinetic energy is $E_k = \frac{1}{2}I\omega^2$, and torque also equals $\tau = Fr \sin\theta$ for a force applied at perpendicular distance r from the axis.

Worked example: follow the reasoning

A uniform disc of mass 2.0 kg and radius 0.30 m ($I = \frac{1}{2}MR^2 = 0.090 \text{ kg}\cdot\text{m}^2$) starts from rest and is spun up by a constant torque of 0.60 N·m for 5.0 s. Find the final angular velocity and the rotational kinetic energy.

$$\alpha = \tau/I = 0.60/0.090 = 6.67 \text{ rad}\cdot\text{s}^{-2}.$$

$$\omega = \alpha t = 6.67 \times 5.0 = 33.3 \text{ rad}\cdot\text{s}^{-1}.$$

$$E_k = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.090)(33.3^2) = 50.0 \text{ J}.$$

Answer: $\omega \approx 33.3 \text{ rad}\cdot\text{s}^{-1}$, $E_k \approx 50 \text{ J}$.

Quick check · multiple choice

A skater spinning with arms extended pulls her arms in, reducing her moment of inertia to half its original value. What happens to her angular velocity, assuming no external torque?

- A Halves
- B Stays the same
- C Doubles
- D Quadruples

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A uniform rod of mass 1.2 kg and length 0.80 m rotates about a perpendicular axis through one end. Its moment of inertia about this axis is $I = \frac{1}{3}mL^2$. [6]
 - (a) Calculate the moment of inertia of the rod.
 - (b) A torque of 0.50 N·m is applied, starting the rod from rest. Calculate the angular acceleration.
 - (c) Calculate the angular velocity after the rod has rotated through 2.0 complete revolutions.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $I = \frac{1}{3}(1.2)(0.80^2) = \frac{1}{3}(1.2)(0.64) = 0.256 \text{ kg}\cdot\text{m}^2$.

(b) $\alpha = \tau/I = 0.50/0.256 = 1.95 \text{ rad}\cdot\text{s}^{-2}$.

(c) $\theta = 2 \times 2\pi = 12.57 \text{ rad}$. $\omega^2 = 2\alpha\theta = 2(1.95)(12.57) = 49.0 \rightarrow \omega = 7.00 \text{ rad}\cdot\text{s}^{-1}$.

Quick check answer

C. Angular momentum $L = I\omega$ is conserved; if I halves, ω must double to keep L constant.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.